

LA HW

Included exercises: 1, 5, 7, 11, 13, 19, 25, 27, 29, 31, 37, 39, 41, 43, 45, 51, 53, 55

Exercises 1–12: Matrix Computations

For Exercises 1–12, compute the indicated matrices, where

$$A = \begin{bmatrix} 2 & -1 & 5 \\ 3 & 4 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 3 & 4 \end{bmatrix}.$$

Original Exercise 1 Compute $4A$.**Original Exercise 5** Compute $(2B)^T$.**Original Exercise 7** Compute $A + B$.**Original Exercise 11** Compute $-(B^T)$.**Exercises 13–24: Compute If Possible**

For Exercises 13–24, compute the indicated matrices, if possible, where

$$A = \begin{bmatrix} 3 & -1 & 2 & 4 \\ 1 & 5 & -6 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} -4 & 0 \\ 2 & 5 \\ -1 & -3 \\ 0 & 2 \end{bmatrix}.$$

Original Exercise 13 Compute $-A$.**Original Exercise 19** Compute $A^T - B$, if possible.**Exercises 25–28: Matrix Entries and Columns**

For Exercises 25–28, assume that

$$A = \begin{bmatrix} 3 & -2 \\ 0 & 1.6 \\ 2\pi & 5 \end{bmatrix}.$$

Original Exercise 25 Determine a_{12} .**Original Exercise 27** Determine $\mathbf{a}_1$, the first column of A .

Exercises 29–32: Rows and Columns

For Exercises 29–32, assume that

$$C = \begin{bmatrix} 2 & -3 & 0.4 \\ 2e & 12 & 0 \end{bmatrix}.$$

Original Exercise 29 Determine c_1 , the first column of C .

Original Exercise 31 Determine the first row of C .

Exercises 37–56: True or False

For Exercises 37–56, determine whether each statement is true or false. Give a brief justification.

Original Exercise 37 Matrices must be of the same size for their sum to be defined.

Original Exercise 39 Every vector is a matrix.

Original Exercise 41 The transpose of a matrix is a matrix of the same size.

Original Exercise 43 If B is a 3×4 matrix, then its rows are 4×1 vectors.

Original Exercise 45 In a zero matrix, every entry is 0.

Original Exercise 51 In any matrix A , the sum of the entries of $3A$ equals three times the sum of the entries of A .

Original Exercise 53 Matrix addition is associative.

Original Exercise 55 If A is a matrix, then cA is the same size as A for every scalar c .